



Chinese Remainder Theorem (CRT)

The Chinese Remainder Theorem (CRT) is a mathematical theorem that provides a solution to a system of congruences. It allows finding a unique solution to a set of simultaneous congruences when the moduli are pairwise coprime (i.e., they have no common factors).

The CRT states that if we have a system of congruences of the form:

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

...

$$x \equiv a_n \pmod{m_n}$$

where $m_1, m_2, \dots, m_n$ are pairwise coprime integers and $a_1, a_2, \dots, a_n$ are any integers, then there exists a unique solution for x modulo M , where M is the product of all the moduli ($M = m_1 * m_2 * \dots * m_n$). Moreover, this solution x can be represented in the form:

$$x \equiv b \pmod{M}$$

The theorem provides an algorithm for finding the value of b , which is called the Chinese Remainder Theorem algorithm. This algorithm involves computing a set of coefficients called the Chinese Remainder Theorem coefficients. These coefficients can be used to find the value of b .

The Chinese Remainder Theorem has various applications in number theory, cryptography, and computer science. It is commonly used in the field of cryptography for efficient modular arithmetic operations. It also has applications in solving linear congruences, solving systems of linear equations, and finding solutions to certain types of Diophantine equations.

Let's consider an example to illustrate the Chinese Remainder Theorem.

Suppose we have the following system of congruences:

$$x \equiv 2 \pmod{3}$$

$$x \equiv 3 \pmod{5}$$

$$x \equiv 2 \pmod{7}$$

To find the solution using the Chinese Remainder Theorem, we first calculate the value of M , which is the product of all the moduli: $M = 3 * 5 * 7 = 105$.

Now, we find the Chinese Remainder Theorem coefficients for each modulus:

For the modulus 3:

$$M_1 = M / m_1 = 105 / 3 = 35$$

$$b_1 \equiv M_1^{(-1)} \pmod{m_1} \equiv 35^{(-1)} \pmod{3} \equiv 2 \pmod{3}$$

For the modulus 5:

$$M_2 = M / m_2 = 105 / 5 = 21$$

$$b_2 \equiv M_2^{(-1)} \pmod{m_2} \equiv 21^{(-1)} \pmod{5} \equiv 1 \pmod{5}$$

For the modulus 7:

$$M_3 = M / m_3 = 105 / 7 = 15$$

$$b_3 \equiv M_3^{(-1)} \pmod{m_3} \equiv 15^{(-1)} \pmod{7} \equiv 1 \pmod{7}$$

Now we can compute the solution x by combining the coefficients and the corresponding remainders:

$$x \equiv a_1 * M_1 * b_1 + a_2 * M_2 * b_2 + a_3 * M_3 * b_3 \pmod{M}$$

$$\equiv 2 * 35 * 2 + 3 * 21 * 1 + 2 * 15 * 1 \pmod{105}$$

$$\equiv 140 + 63 + 30 \pmod{105}$$

$$\equiv 233 \pmod{105}$$

Therefore, the solution to the system of congruences is $x \equiv 233 \pmod{105}$.

Note that this solution is unique modulo 105, but there are infinitely many solutions that differ by multiples of 105. In this case, any value of x that is congruent to 233 modulo 105 is a valid solution.

```

1. import java.util.*;
2. class ChineseRemainder {
3.     int calculate(int size , int
    div[], int rem[])
4.     {
5.         int j , x = 1;
6.         while(true)
7.         {
8.             for( j=0;j<size;j++)
9.             {
10.                if( x % div[j] !=
    rem[j] )
11.                break;
12.            }
13.            if(j == size)
14.                return x;
15.            x++;} }}

16.class ChineseRemainderTheorem
17.{
18.    public static void main (String[] args)
19.    {

```

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1. Scanner sc=new Scanner(System.in);
2.     System.out.println("Enter
    Diviser");
3.     int size = sc.nextInt();
4.     int [] div = new int[size];
5.     for(int i=0; i<n; i++)
6.     {
7.         div[i] = sc.nextInt();
8.     }
9.
10.    System.out.println("Enter
    Remainder");
11.    int rem[]=new int[size];
12.    for(int i=0; i<size; i++)
13.    {
14.        rem[i] = sc.nextInt();
15.    }
16.
17.    ChineseRemainder  c = new
    ChineseRemainder();
18.    System.out.println( c.calculate(size,
    div, rem));
19.    } }

```



THANK YOU